This assignment will not be graded and is only for practice.

Note: Suppose W_1 **and** W_2 are subspaces of a vector space V. Then sum of two subspaces is defined as:

$$W_1 + W_2 = \{w_1 + w_2 \mid w_1 \in W_1, w_2 \in W_2\}.$$

One can verify that $W_1 + W_2$ is a subspace of V.

Multiple Choice Questions (MCQ):

1) An inner product on R^3 is defined as: **1 point**

$$\langle .,. \rangle : \mathrm{R}^3 \times \mathrm{R}^3 \rightarrow \mathrm{R}$$

$$\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle = x_1 y_1 + x_2 y_2 + x_3 y_3.$$

Match a set S in Column A with the set T given in column B, so that $T \subset Span(S)^{\perp}$. Match a set in column B with the property it satisfies in column C.

	Set (S) (Column A)		set (T) (Column B)		Properties (Column C)
a)	$\{(-2,1,3)\}$	i)	$\{(0,1,1),(2,-1,1)\}$	1)	forms an orthonormal basis of a vector space which is isomorphic to $\mathbb{R}^2$
b)	$\{(1,-2,1)\}$	ii)	$\{(1,-1,1)(2,1,1)\}$	2)	orthogonal but not orthonormal
c)	$\{(1,1,-1)\}$	iii)	$\{\frac{1}{\sqrt{2}}\ (0,-1,1)\ \}$	3)	not orthogonal
d)	$\{(0,1,1)(1,1,1)\}$	iv)	$\{\frac{1}{\sqrt{3}}(1,1,1), \frac{1}{\sqrt{2}}(-1,0,1)\}$	4)	orthonormal but does not form a basis of a vector space which is isomorphic to $\mathbb{R}^2$

Table : M2W9P1

Which of the following option are true?

- [ ] d $\rightarrow$ ii $\rightarrow$ 3
- [ ] d $\rightarrow$ iii $\rightarrow$ 4
- [ ] c $\rightarrow$ ii $\rightarrow$ 4
- [ ]

a → iv → 1

- [ ] a → ii → 3

- [ ] b → iv → 2

No, the answer is incorrect.
Score: 0

Accepted Answers:

d→ iii → 4
a → ii → 3

2) Suppose W_1 and W_2 are subspaces of a vector space V. Let P_{W_1} and P_{W_2} denote the projection from V to W_1 and V to W_2 respectively and consider the following statements:
Statement P: If $P_{W_1} + P_{W_2}$ is a projection from V to $W_1 + W_2$, then $P_{W_1} \circ P_{W_2} + P_{W_2} \circ P_{W_1} = 0$.

Statement Q: $P^2_{W_1} := P_{W_1} \circ P_{W_1}$ is not a projection from V to W_1.

Statement R: If A is the matrix representation of P_{W_2}, then A can not be a symmetric matrix.

Statement S: $P_{W_1} - P_{W_2}$ is a projection from V to $W_1 + W_2$.

Find the number of correct statements

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

3) Which of the followings option(s) is (are) true? ***1 point***

- [ ] Let $u, v,$ and w be three vectors in $\mathbb{R}^2$ and $\langle ., . \rangle$ be an inner product on $\mathbb{R}^2$. If $\langle u, v \rangle = \langle u, w \rangle$, then $v = w$

- [ ] Let V be an inner product space. If $\langle u, v \rangle = 0$ for all $u \in V$, then $v = 0$

- [ ] Let V be an inner product space and $u, v \in V$. If $\langle u+v, u-v \rangle = 0$, then $\|u\| = \|v\|$
- [ ] Let V be an inner product space. If $S \subset V$ then $S^{\perp} = (span(S))^{\perp}$
- [ ] Suppose $\{u_1, u_2, \dots u_m\}$ be an orthogonal set of vectors in an inner product space V, then $\|u_1 + u_2 + \dots + u_m\|^2 = \|u_1\|^2 + \|u_2\|^2 + \dots + \|u_m\|^2$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Let V be an inner product space. If $\langle u, v \rangle = 0$ for all $u \in V$, then $v = 0$
Let V be an inner product space. If $S \subset V$ then $S^{\perp} = (span(S))^{\perp}$
Suppose $\{u_1, u_2, \dots u_m\}$ be an orthogonal set of vectors in an inner product space V, then $\|u_1 + u_2 + \dots + u_m\|^2 = \|u_1\|^2 + \|u_2\|^2 + \dots + \|u_m\|^2$.

4) Which of the following option(s) is/are true? ***1 point***

- [ ] Let A and B be orthogonal matrices of order 3. Then A and B are equivalent matrices.
- [ ] Let A and B be orthogonal matrices. Then A and B are similar matrices.
- [ ] Let A be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle θ. Then nullity of the matrix A is 0.
- [ ] Let A be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle θ and B be a rotation matrix corresponding to the anti- clock wise rotation of the YZ-plane about the X - axis with angle β. Then A and B are equivalent matrices.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Let A and B be orthogonal matrices of order 3. Then A and B are equivalent matrices.
Let A be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle θ. Then nullity of the matrix A is 0.
Let A be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle θ and B be a rotation matrix corresponding to the anti- clock wise

rotation of the YZ-plane about the X -axis with angle β. Then A and B are equivalent matrices.

Numerical Answer Type:

5) Let $u = (1, 2, 1)$ and $v = (1, 2, 3)$ be the vectors of the inner product space $\mathbb{R}^3$ with usual inner product. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be an orthogonal linear transformation. If θ is an angle between $T(u)$ and $T(v)$, then find the value of $||T(u)||\,||T(v)||\cos\theta$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 8

1 point

6) Consider a set $W = \{(1, 1, 1)\}$ from the inner product space $\mathbb{R}^3$. Find the dimension of the subspace $W^{\perp}$, where $W^{\perp}$ is the collection of the vectors which are orthogonal to the vector (1,1,1).

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

7) Let $v = (3, 1, 2)$ be a vector in $\mathbb{R}^3$. If (a, b, c) is the vector obtained from v after the anti-clock wise rotation of ZX-plane with angle 45° about the Y- axis, then find the value of $\sqrt{2}(a + b + c - 1)$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 6

1 point

Comprehension Type Question:

With a particular frame of reference (in R^2) the position of Rahul's house is at the point (8,1). There are two roads Road 1 and Road 2 along the lines $y = 2x \text{ and } y = -2x$ respectively on the $XY-$ plane.
Answer questions 8,9 and 10 using the given information.

8) Which of the following option(s) is (are) true? **1 point**

- [ ] If Rahul wants to travel the minimum distance from his house to Road 1, then he will meet Road 1 at (2,4).
- [ ] If Rahul wants to travel the minimum distance from his house to Road 1, then he will meet Road 1 at $\frac{6}{5}(1,-2)$.
- [ ] If Rahul wants to travel the minimum distance from his house to Road 2, then he will meet Road 2 at $\frac{6}{5}(1,-2)$.
- [ ] If Rahul wants to travel the minimum distance from his house to Road 2, then he will meet Road 2 at (2,4).

No, the answer is incorrect.

Score: 0

Accepted Answers:

If Rahul wants to travel the minimum distance from his house to Road 1, then he will meet Road 1 at (2,4).
If Rahul wants to travel the minimum distance from his house to Road 2, then he will meet Road 2 at $\frac{6}{5}(1,-2)$.

9) Let θ be the acute angle between Road 1 and Road 2, then what will be the value of $5\cos(\theta)$?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

10) Let $\langle .,. \rangle$ denote the standard inner product on $\mathbb{R}^2$, i.e., $\langle (x_1,x_2),(y_1,y_2)\rangle = x_1y_1 + x_2y_2$. Suppose $W_1 = \{(x,y) \mid y = 2x\}$ and $W_2 = \{(x,y) \mid y = -2x\}$ are two subspaces of $\mathbb{R}^2$. Which of the following options is correct? **1 point**

- [ ] $W_1^{\perp} = \{(x,y) \mid 2y = x\}$

- [ ] $W_2^{\perp} = \{(x, y) \mid 2y = -x\}$
- [ ] $W_1^{\perp} = \{(x, y) \mid 2y = -x\}$
- [ ] $W_2^{\perp} = \{(x, y) \mid 2y = x\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$W_1^{\perp} = \{(x, y) \mid 2y = -x\}$
$W_2^{\perp} = \{(x, y) \mid 2y = x\}$

Your score is: 0/10